KAIZEN MATHS

Lesson starter

Sampling Methods and Bias

Methods & Bias

Name _____

S1

Name the sampling method used in each scenario.

Name the sampling method.A manager lists 1200 customer records in order and selects every 30th customer after a random starting point.

S2

Identify the source of bias and explain why it matters.

Identify the likely source of bias and explain why it is a problem.A survey asks, 'Do you agree that the excellent new park should stay open late?'

S3

Suggest an improvement to make each sampling method more representative.

Suggest one improvement to make the sampling method more representative.A transport survey is carried out only at a train station at 8:00 am.

S4

Identify the problem with each questionnaire item and rewrite it neutrally.

Explain what is wrong with the question and write a better version."Do you exercise regularly and eat healthily?"

KAIZEN MATHS

Lesson starter

Sampling Methods and Bias

Methods & Bias

Name _____

S1

Name the sampling method used in each scenario.

Name the sampling method.A manager lists 1200 customer records in order and selects every 30th customer after a random starting point.

S2

Identify the source of bias and explain why it matters.

Identify the likely source of bias and explain why it is a problem.A survey asks, 'Do you agree that the excellent new park should stay open late?'

S3

Suggest an improvement to make each sampling method more representative.

Suggest one improvement to make the sampling method more representative.A transport survey is carried out only at a train station at 8:00 am.

S4

Identify the problem with each questionnaire item and rewrite it neutrally.

Explain what is wrong with the question and write a better version."Do you exercise regularly and eat healthily?"

STA-8B36-L1 · S

Show your working

KAIZEN MATHS

Exit ticket

Sampling Methods and Bias

Methods & Bias

Name _____

EA1

Name the sampling method used in each scenario.

Name the sampling method.A website posts a link inviting readers to complete a survey if they want to take part.

EA2

Identify the source of bias and explain why it matters.

Identify the likely source of bias and explain why it is a problem.A company emails 2000 customers, but only 90 reply.

Show your working

KAIZEN MATHS

Exit ticket

Sampling Methods and Bias

Methods & Bias

Name _____

EB1

Suggest an improvement to make each sampling method more representative.

Suggest one improvement to make the sampling method more representative.A council surveys residents using only landline telephone numbers.

Show your working

KAIZEN MATHS

Exit ticket

Sampling Methods and Bias

Methods & Bias

Name _____

EA1

Name the sampling method used in each scenario.

Name the sampling method.A website posts a link inviting readers to complete a survey if they want to take part.

EA2

Identify the source of bias and explain why it matters.

Identify the likely source of bias and explain why it is a problem.A company emails 2000 customers, but only 90 reply.

STA-8B36-L1 · EA

Show your working

KAIZEN MATHS

Exit ticket

Sampling Methods and Bias

Methods & Bias

Name _____

EB1

Suggest an improvement to make each sampling method more representative.

Suggest one improvement to make the sampling method more representative.A council surveys residents using only landline telephone numbers.

STA-8B36-L1 · EB

Show your working

Teacher answers & working

Sampling Methods and Bias · Methods & Bias

STA-8B36-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Name the sampling method used in each scenario.

Name the sampling method.A manager lists 1200 customer records in order and selects every 30th customer after a random starting point.

Answer:

Systematic Sampling

Working:

Look at how the people or items are selected.

Key evidence: The sample is chosen at a regular interval from an ordered list.

The method is systematic sampling.

S2

Identify the source of bias and explain why it matters.

Identify the likely source of bias and explain why it is a problem.A survey asks, 'Do you agree that the excellent new park should stay open late?'

Answer:

Question Wording Bias: The word 'excellent' pushes respondents towards a positive answer.

Working:

First identify who can be selected or who is likely to respond.

In this case: The word 'excellent' pushes respondents towards a positive answer.

This is question wording bias because the sample is not likely to represent the full population fairly.

A better approach: Use neutral wording, such as 'Should the new park stay open late?'

S3

Suggest an improvement to make each sampling method more representative.

Suggest one improvement to make the sampling method more representative.A transport survey is carried out only at a train station at 8:00 am.

Answer:

Use several locations and times, then select people randomly from the wider passenger population.

Working:

A good improvement should address the source of bias, not just increase the sample size.

Problem with the method: People at a train station at 8:00 am are likely to be commuters, so other travellers are under-represented.

Improvement: Use several locations and times, then select people randomly from the wider passenger population.

This improves representativeness because more of the target population has a fair chance of being included.

Teacher answers & working

Sampling Methods and Bias · Methods & Bias

STA-8B36-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S4

Identify the problem with each questionnaire item and rewrite it neutrally.

Explain what is wrong with the question and write a better version."Do you exercise regularly and eat healthily?"

Answer:

Problem: It asks two different things at once. Better version: "How often do you exercise?"

Working:

Check whether the question is leading, double-barrelled, unclear, or loaded with opinion words.

Problem: It asks two different things at once.

A neutral version is: "How often do you exercise?"

The improved question is more likely to collect valid responses.

EA1

Name the sampling method used in each scenario.

Name the sampling method.A website posts a link inviting readers to complete a survey if they want to take part.

Answer:

Volunteer Sampling

Working:

Look at how the people or items are selected.

Key evidence: People decide for themselves whether to join the sample.

The method is volunteer sampling.

EA2

Identify the source of bias and explain why it matters.

Identify the likely source of bias and explain why it is a problem.A company emails 2000 customers, but only 90 reply.

Answer:

Non-Response Bias: The people who reply may differ from those who do not reply.

Working:

First identify who can be selected or who is likely to respond.

In this case: The people who reply may differ from those who do not reply.

This is non-response bias because the sample is not likely to represent the full population fairly.

A better approach: Use reminders, offer multiple response methods, and compare responders with the full customer list.

Teacher answers & working

Sampling Methods and Bias · Methods & Bias

STA-8B36-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

EB1

Suggest an improvement to make each sampling method more representative.

Suggest one improvement to make the sampling method more representative.A council surveys residents using only landline telephone numbers.

Answer:

Use a broader sampling frame such as an address list or electoral register.

Working:

A good improvement should address the source of bias, not just increase the sample size.

Problem with the method: Residents without landlines cannot be selected, so the sampling frame misses part of the population.

Improvement: Use a broader sampling frame such as an address list or electoral register.

This improves representativeness because more of the target population has a fair chance of being included.